

MATH 2790 Final Exam Fall 2025

Name: _____ Auburn ID: _____

Instructions

1. Print your full name and your banner ID on your exam. Then finish reading these instructions, and sign the bottom of this page.
2. All electronic equipment (cell phones, Apple Watch, AirPods, etc) are **NOT ALLOWED** during the exam time.
3. You are allowed to use up to 2 calculators from the following list: BA-35, BA II Plus, BA II Plus Professional, TI-30Xa, TI-30X II (IIS solar or IIB battery), TI-30XS MultiView (or XB battery).
4. No talking is allowed during the exam.
5. Unless otherwise indicated, **SHOW ALL YOUR WORK**. If no work is shown, no partial credit can be awarded. Your work and answer need to be accurate and relevant to receive points.
6. You will be given exactly 120 minutes for this exam.
7. Any student not following the above instructions nor behaving according to the above instructions during the exam may have their exam confiscated and points deducted.

I have read and fully understand all of the above instructions.

Signature: _____

Do not write in the table below

Problem	Points	Score
1	11	
2	11	
3	11	
4	11	
5	11	
6	11	
7	11	
8	11	
9	11	
10	11	
Total	110	

Your grade will be scored out of 100. If you earned X out of 110 points, you will receive an $X\%$.

1. (11 points) A 15-year bond is purchased at a premium. The bond pays semiannual coupons. The amount of amortization of premium in the 6th coupon is 80.00. The amount of amortization of premium in the 27th coupon is 148.82. Calculate the premium of this bond. Round to 2 decimal places.

- 2. (11 points)** Given that the force of interest is $\delta_t = 0.02 + 0.01t$,
- (a) (9 points) Find the accumulation function $a(t)$.
 - (b) (6 points) Find the value at time $t = 2$ of \$300 to be paid at time $t = 5$. Round to 2 decimal places.

3. (11 points) A small business owner, Maria, borrows \$25,000 from a local credit union to upgrade her equipment. Maria repays the loan by making a payment of \$10,000 at the end of one year, \$9,000 at the end of two years, and \$8,500 at the end of four years. The money received by the credit union is immediately reinvested at an annual effective rate of 2% from years 1 to 2, and then at 3.5% thereafter.

- (a) (8 points) Write out the equation to solve for Maria's annual effective interest rate on the loan. Use the financial calculator to find the value. Round to 4 decimal places as a percentage.
- (b) (8 points) Calculate the credit union's annual yield rate on this transaction. Round to 4 decimal places as a percentage.

4. (11 points) An athlete, Emma, signs a sponsorship deal. She receives \$4 at the end of the first month, \$7 at the end of the second month, and thereafter each monthly payment increases by \$3. The amount of the last payment is \$64.

If for the first year, the nominal interest rate is 6% convertible monthly and then increases to 9% convertible monthly. Calculate the value of Emma's sponsorship deal at the time of her first payment.

5. (11 points) Lucy is awarded a scholarship that will pay her \$800 at the end of the first year. Each subsequent yearly payment increases by 1%. The effective annual interest rate is 5%.

- (a) Suppose the scholarship payments continue for 10 years. Calculate the value of these payments at the time of the last payment.
- (b) Now suppose instead that the scholarship payments continue forever. Lucy plans to reevaluate her finances at the time of her third payment. Calculate the value, at that time, of all scholarship payments (both past and future).

6. (11 points) A loan of \$30,000 is taken out at an annual effective interest rate of 2%. Jordan makes annual payments that are five times the amount of interest accrued in the year. He continues this process until he can make one final payment of at most \$250. Find the time and the amount of Jordan's final payment. Round to 2 decimal places.

7. (11 points) For the next 15 years, Rosie will deposit an amount of \$1000 at the end of each month into an account that earns a nominal interest rate of 6%, compounded monthly. Immediately after her last monthly deposit, the accumulated balance is transferred into a scholarship fund that instead earns an annual effective interest rate of 4%. How large of a yearly scholarship can Rosie give out forever, assuming that all of the deposited amount goes toward the scholarship and the first scholarship is given one year after the account is opened?

8. (11 points) Bond A is an 8-year \$1,400 par-value bond with a nominal coupon rate of 8%, convertible semiannually.

Bond B is a \$1,000 bond with a nominal coupon rate of 12%, convertible semiannually, and a redemption amount of \$1,137.80.

If both bonds yield a nominal rate of 6% convertible semiannually and have the same price, how long (in years) until Bond B matures?

9. (11 points) Account A is funded by an annuity-due with 10 payments of \$120 and earns an annual effective interest rate of 5%. Account B is funded with a single deposit of \$1,200 today and earns simple interest at a rate of 3% per year. Which account has a larger accumulated value at the end of 10 years? Justify your answer.

10. (11 points) A car is purchased for \$32,000. A down payment of \$2,000 is made at the time of purchase, and the remaining balance is financed with a loan to be repaid with level monthly payments over 6 years. The loan is subject to a nominal *discount* rate of 4.8%, convertible monthly.

- (a) Calculate the amount of the monthly payment. *Round up* to 2 decimals.
- (b) Calculate the total interest paid during the first 3 years of the loan assuming the payments found in (a) are made every month except for a reduced final payment.

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